











































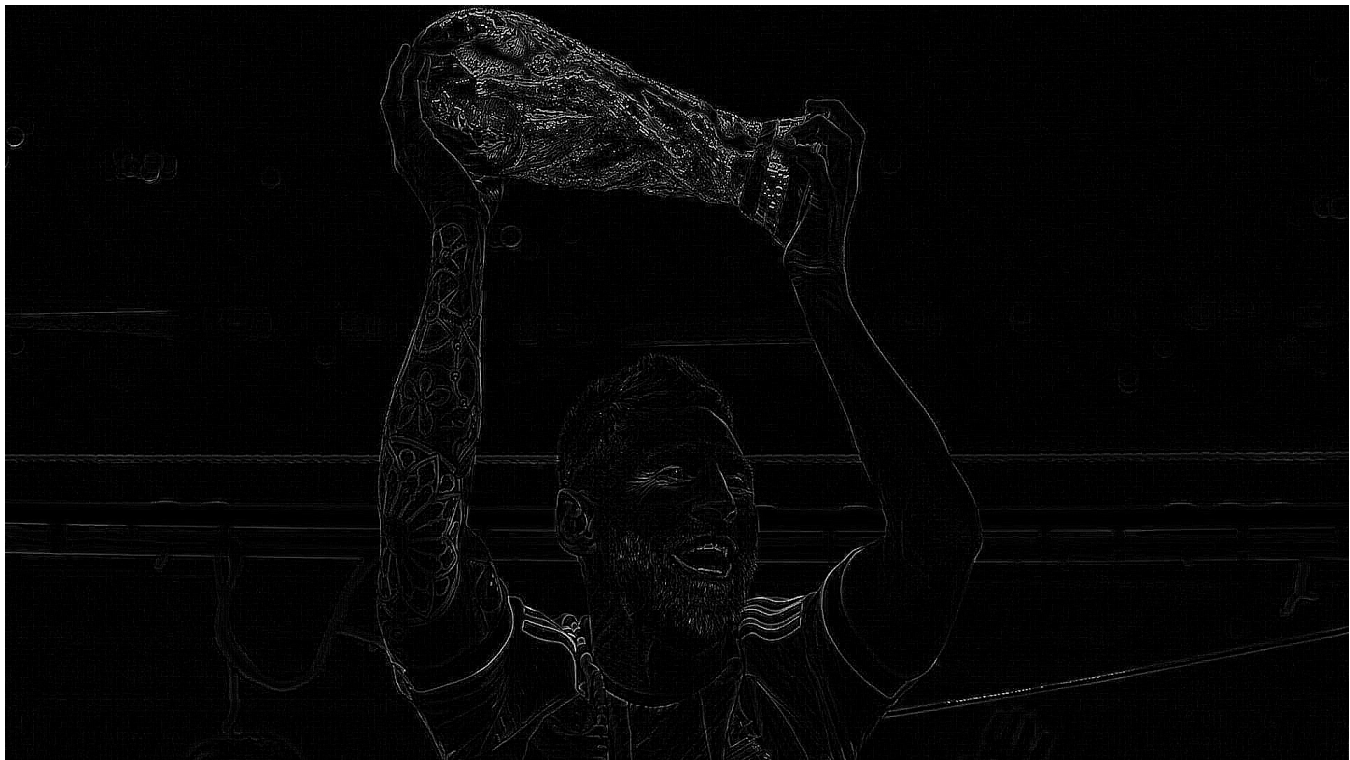




































DSA0205-computer-vision-with-open-cv-main/Exp 1.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

cv2.imshow('Original', img)
cv2.imshow('Grayscale', gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 2.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
blur = cv2.GaussianBlur(img, (7, 7), 0)

cv2.imshow('Original', img)
cv2.imshow('Blurred', blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 3.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)

cv2.imshow('Original', img)
cv2.imshow('Outline (Canny)', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 4.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
kernel = np.ones((5, 5), np.uint8)
dilated = cv2.dilate(img, kernel, iterations=1)

cv2.imshow('Original', img)
cv2.imshow('Dilated', dilated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 5.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
kernel = np.ones((5, 5), np.uint8)
eroded = cv2.erode(img, kernel, iterations=1)

cv2.imshow('Original', img)
cv2.imshow('Eroded', eroded)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 6.txt

Text preview 1/1

```
import cv2

cap = cv2.VideoCapture('video.mp4')
fps = cap.get(cv2.CAP_PROP_FPS)

mode = input("Enter mode - slow/fast: ").strip().lower()
delay = int(1000 / fps)
delay = delay * 3 if mode == 'slow' else max(1, delay // 3)

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow('Video Playback', frame)
    if cv2.waitKey(delay) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 7.txt

Text preview 1/1

```
import cv2

cap = cv2.VideoCapture(0)
mode = input("Enter mode - slow/fast: ").strip().lower()
delay = 100 if mode == 'slow' else 5

while True:
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow('Webcam', frame)
    if cv2.waitKey(delay) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 8.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
bigger = cv2.resize(img, None, fx=1.5, fy=1.5, interpolation=cv2.INTER_LINEAR)
smaller = cv2.resize(img, None, fx=0.5, fy=0.5, interpolation=cv2.INTER_AREA)

cv2.imshow('Original', img)
cv2.imshow('Bigger', bigger)
cv2.imshow('Smaller', smaller)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 9.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
(h, w) = img.shape[:2]
center = (w // 2, h // 2)

M_cw = cv2.getRotationMatrix2D(center, -45, 1.0)
M_ccw = cv2.getRotationMatrix2D(center, 45, 1.0)

rotated_cw = cv2.warpAffine(img, M_cw, (w, h))
rotated_ccw = cv2.warpAffine(img, M_ccw, (w, h))

cv2.imshow('Clockwise', rotated_cw)
cv2.imshow('Counter-Clockwise', rotated_ccw)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 10.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
(h, w) = img.shape[:2]

M = np.float32([[1, 0, 100], [0, 1, 50]]) # shift x=100, y=50
shifted = cv2.warpAffine(img, M, (w, h))

cv2.imshow('Original', img)
cv2.imshow('Translated', shifted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 11.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
rows, cols = img.shape[:2]

pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

M = cv2.getAffineTransform(pts1, pts2)
result = cv2.warpAffine(img, M, (cols, rows))

cv2.imshow('Original', img)
cv2.imshow('Affine Transform', result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 12.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
rows, cols = img.shape[:2]

pts1 = np.float32([[56, 65], [368, 52], [28, 387], [389, 390]])
pts2 = np.float32([[0, 0], [300, 0], [0, 300], [300, 300]])

M = cv2.getPerspectiveTransform(pts1, pts2)
result = cv2.warpPerspective(img, M, (300, 300))

cv2.imshow('Original', img)
cv2.imshow('Perspective Transform', result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 13.txt

Text preview 1/1

```
import cv2
import numpy as np

cap = cv2.VideoCapture('video.mp4')
pts1 = np.float32([[56, 65], [368, 52], [28, 387], [389, 390]])
pts2 = np.float32([[0, 0], [300, 0], [0, 300], [300, 300]])
M = cv2.getPerspectiveTransform(pts1, pts2)

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    warped = cv2.warpPerspective(frame, M, (300, 300))
    cv2.imshow('Original', frame)
    cv2.imshow('Perspective Video', warped)
    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 14.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
rows, cols = img.shape[:2]

pts_src = np.array([[141, 131], [480, 159], [493, 630], [64, 601]])
pts_dst = np.array([[0, 0], [400, 0], [400, 600], [0, 600]])

H, status = cv2.findHomography(pts_src, pts_dst)
result = cv2.warpPerspective(img, H, (400, 600))

cv2.imshow('Original', img)
cv2.imshow('Homography Result', result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 15.txt

Text preview 1/1

```
import cv2
import numpy as np

def compute_dlt(pts_src, pts_dst):
    A = []
    for i in range(len(pts_src)):
        x, y = pts_src[i][0], pts_src[i][1]
        u, v = pts_dst[i][0], pts_dst[i][1]
        A.append([-x, -y, -1, 0, 0, 0, x*u, y*u, u])
        A.append([0, 0, 0, -x, -y, -1, x*v, y*v, v])
    A = np.array(A)
    _, _, Vt = np.linalg.svd(A)
    H = Vt[-1].reshape(3, 3)
    return H / H[2, 2]

img = cv2.imread('image.jpg')
pts_src = np.array([[141, 131], [480, 159], [493, 630], [64, 601]], dtype=np.float32)
pts_dst = np.array([[0, 0], [400, 0], [400, 600], [0, 600]], dtype=np.float32)

H = compute_dlt(pts_src, pts_dst)
result = cv2.warpPerspective(img, H, (400, 600))

cv2.imshow('Original', img)
cv2.imshow('DLT Result', result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 16.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)

cv2.imshow('Canny Edges', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 17.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobelx = cv2.convertScaleAbs(sobelx)

cv2.imshow('Sobel X', sobelx)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 18.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobely = cv2.convertScaleAbs(sobely)

cv2.imshow('Sobel Y', sobely)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 19.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobelxy = cv2.magnitude(sobelx, sobely)
sobelxy = cv2.convertScaleAbs(sobelxy)

cv2.imshow('Sobel XY', sobelxy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 20.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[0, 1, 0],
                   [1, -4, 1],
                   [0, 1, 0]])

laplacian = cv2.filter2D(gray, -1, kernel)
sharpened = cv2.subtract(gray, laplacian)

cv2.imshow('Sharpened (Negative Center)', sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 21.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[1, 1, 1],
                   [1, -8, 1],
                   [1, 1, 1]])

laplacian = cv2.filter2D(gray, -1, kernel)
sharpened = cv2.subtract(gray, laplacian)

cv2.imshow('Sharpened (Diagonal Extension)', sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 22.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[1, 1, 1],
                   [1, -8, 1],
                   [1, 1, 1]])

laplacian = cv2.filter2D(gray, -1, kernel)
sharpened = cv2.subtract(gray, laplacian)

cv2.imshow('Sharpened (Diagonal Extension)', sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 23.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
blurred = cv2.GaussianBlur(img, (9, 9), 10.0)
unsharp = cv2.addWeighted(img, 1.5, blurred, -0.5, 0)

cv2.imshow('Original', img)
cv2.imshow('Unsharp Masking', unsharp)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 24.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
blurred = cv2.GaussianBlur(img, (9, 9), 10.0)
mask = cv2.subtract(img, blurred)

A = 1.5 # boost factor ( $A > 1$ )
high_boost = cv2.addWeighted(img, A, mask, 1, 0)

cv2.imshow('High-Boost Result', high_boost)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 25.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
gradient = cv2.convertScaleAbs(cv2.magnitude(sobelx, sobely))

sharpened = cv2.addWeighted(gray, 1.0, gradient, 0.5, 0)

cv2.imshow('Gradient Sharpened', sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 26.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
h, w = img.shape[:2]

watermark_text = "SIMATS"
cv2.putText(img, watermark_text, (w - 200, h - 20),
            cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 255), 2, cv2.LINE_AA)

cv2.imshow('Watermarked Image', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 27.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')
background = cv2.imread('background.jpg')

# Crop a region
cropped = img[50:200, 50:200]

# Paste into background at position (100, 100)
background[100:100+cropped.shape[0], 100:100+cropped.shape[1]] = cropped

cv2.imshow('Cropped', cropped)
cv2.imshow('Pasted Result', background)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 28.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[ -1, -1, -1],
                   [-1,  8, -1],
                   [-1, -1, -1]])

boundary = cv2.filter2D(gray, -1, kernel)

cv2.imshow('Boundary Detection', boundary)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 29.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((5, 5), np.uint8)
erosion = cv2.erode(img, kernel, iterations=1)

cv2.imshow('Erosion', erosion)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 30.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((5, 5), np.uint8)
dilation = cv2.dilate(img, kernel, iterations=1)

cv2.imshow('Dilation', dilation)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 31.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((5, 5), np.uint8)
opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)

cv2.imshow('Opening', opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 32.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((5, 5), np.uint8)
closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

cv2.imshow('Closing', closing)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 33.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((5, 5), np.uint8)
gradient = cv2.morphologyEx(img, cv2.MORPH_GRADIENT, kernel)

cv2.imshow('Morphological Gradient', gradient)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 34.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((9, 9), np.uint8)
tophat = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)

cv2.imshow('Top Hat', tophat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 36.txt

Text preview 1/1

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)
kernel = np.ones((9, 9), np.uint8)
blackhat = cv2.morphologyEx(img, cv2.MORPH_BLACKHAT, kernel)

cv2.imshow('Black Hat', blackhat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 37.txt

Text preview 1/1

```
import cv2

cap = cv2.VideoCapture('video.mp4')
frames = []

while True:
    ret, frame = cap.read()
    if not ret:
        break
    frames.append(frame)
cap.release()

for frame in reversed(frames):
    cv2.imshow('Reverse Playback', frame)
    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 38.txt

Text preview 1/1

```
import cv2

face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5)

for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x + w, y + h), (255, 0, 0), 2)

cv2.imshow('Face Detection', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 39.txt

Text preview 1/1

```
import cv2

car_cascade = cv2.CascadeClassifier('cars.xml') # Haar cascade for vehicles

cap = cv2.VideoCapture('video.mp4')

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    cars = car_cascade.detectMultiScale(gray, 1.1, 3)

    for (x, y, w, h) in cars:
        cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 255), 2)

    cv2.imshow('Vehicle Detection', frame)
    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Exp 40.txt

Text preview 1/1

```
import cv2

img = cv2.imread('image.jpg')

# Draw rectangle around region of interest
x, y, w, h = 50, 50, 200, 150
cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 2)

# Extract the object inside the rectangle
extracted = img[y:y+h, x:x+w]

cv2.imshow('Rectangle Drawn', img)
cv2.imshow('Extracted Object', extracted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Read an image in python and Co

Text preview 1/1

```
import cv2
img = cv2.imread(r"C:\Users\prash\Downloads\Computer Vision.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)
cv2.imwrite('Canny_Edges_3rd.jpg', edges)
cv2.imshow('Edges', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Read an image in python and Di

Text preview 1/1

```
import cv2
import numpy as np
img = cv2.imread(
    r"C:\Users\prash\Downloads\Computer Vision.jpg",
    cv2.IMREAD_GRAYSCALE
)
kernel = np.ones((5, 5), np.uint8)
dilated_img = cv2.dilate(img, kernel, iterations=1)
cv2.imwrite('Dilated_Image.jpg', dilated_img)
cv2.imshow("Original", img)
cv2.imshow("Dilated Image", dilated_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

DSA0205-computer-vision-with-open-cv-main/Read an image in python and Er

Text preview 1/1

```
import cv2
import numpy as np
img = cv2.imread(
    r"C:\Users\prash\Downloads\Computer Vision.jpg")
gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))
eroded_image = cv2.erode(gray_image, kernel, iterations=1)
cv2.imwrite("eroded_image.jpg", eroded_image)
cv2.imshow("Original", gray_image)
cv2.imshow("Eroded Image", eroded_image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```